本次任务的目标是：先保证光变测光结果物理可信、能区分噪声、系统效应与真实变异，并先在小而可控的 Seyfert 样本上把流程验证清楚，再逐步扩展到更弱的 AGN 及其他类别。

1.数据读取与 sanity check

1.1 核心源唯一性

本次使用的 MINFIX r-band 光变曲线在 QC 图中均显示 oid_n=1，且只用 r-band，确保不会出现先前的“可能把多个源混在一条曲线里”的风险。

1.2 质量控制

QC 图中显示使用数据均满足：

（1）catflags=[0]

（2）outlier 只做标记，不做自动删点

（3）接近 limitmag 被显式标记，用于识别低 SNR 噪声主导区

交付物：

Seyfert_bright_QC_booklet.pdf

2. Seyfert 控制样本（bright & well-sampled）QC

2.1 样本定义

（1）Mean_mag < 18

（2）N > 500

2.2 代表性光变曲线图册

输出 30 个 bright Seyfert 的 QC 图册，每页含：

（1）光变曲线（含误差棒）

（2）outlier 红圈标记、near-limit 橙圈标记

（3）元信息（Mean、Fvar、N、ΔT）

（4）完整性指标（oid_n、catflags、n_outlier、n_near_limit）

交付物：

① Seyfert_bright_QC_booklet.pdf

② qc_index.csv

3.区分真实变异与噪声

3.1 证据：固定阈值 Fvar>5%不合理

全样本中 Fvar 随 Mean_mag 变暗显著增大，暗端明显噪声、系统误差主导。

交付图：

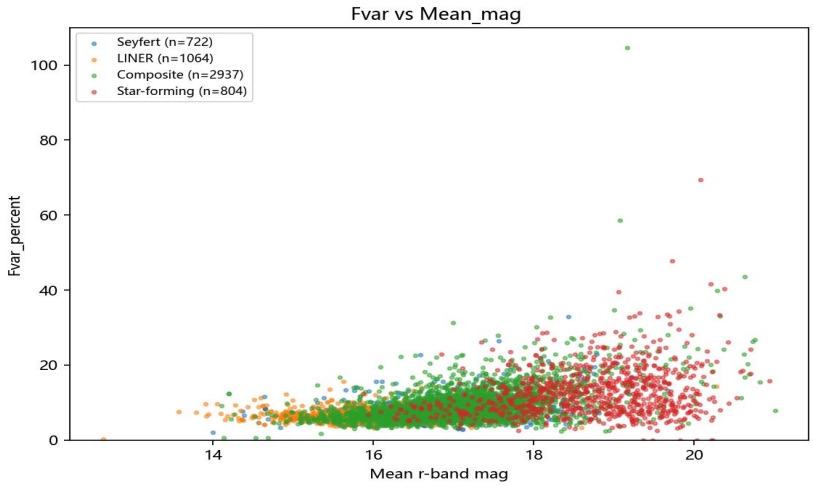

图 1 不同 BPT 类别星系的 Fvar – 平均星等关系（ZTF r 波段）

图 1 展示了 Seyfert、LINER、Composite 与 Star-forming 样本在 ZTF r 波段的分数变异幅度 Fvar_percent 随平均星等 Mean r-band mag 的变化。可见在更暗的源中 Fvar 的离散度与上包络明显升高，提示光度误差、系统效应在暗端占主导，因此采用固定阈值，例如 Fvar>5%，会在暗端产生更高的“虚假可变”比例；后续分析需引入随星等变化的噪声基线或更稳健的显著性指标。

3.2 用 SF 建立“随星等变化”的噪声基线阈值

在每个 Mean_mag 分箱内，用 star-forming 的 Fvar 95%分位数作为阈值，替代固定 cut。

关键表：

SF_thresholds_by_magbin_Fvar.csv

3.3 基于 SF 阈值的可变率对比

输出两张图：

① 全样本四类的可变率 vs Mean_mag bin

② Seyfert GOLD（Mean<18, N>500）版本

交付图：

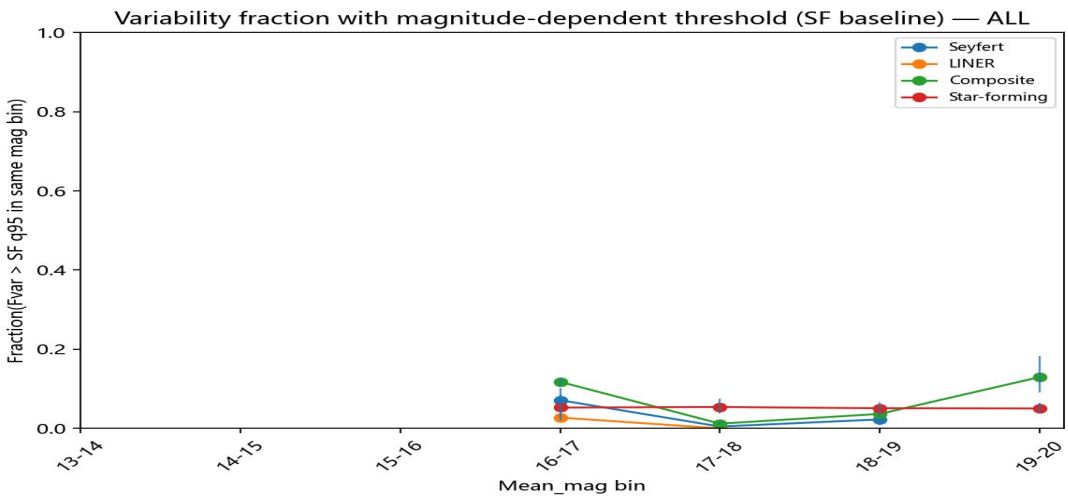

图 2 基于星等分箱的可变率：采用 Star-forming 噪声基线的星等相关阈值（全样本）

图2展示在不同平均星等分箱内，各类别（Seyfert、LINER、Composite、Star-forming）中满足 Fvar > SF_q95 的对象比例。这里 SF_q95 表示 star-forming 样本在该星等分箱内的 Fvar 95%分位数，作为噪声、系统效应的基线阈值。由于阈值由 SF 定义，SF 自身在各分箱的比例应接近 5%。该星等相关阈值用于替代固定阈值，从而缓解暗端噪声抬升导致的误判；误差棒表示分箱内比例的不确定度。

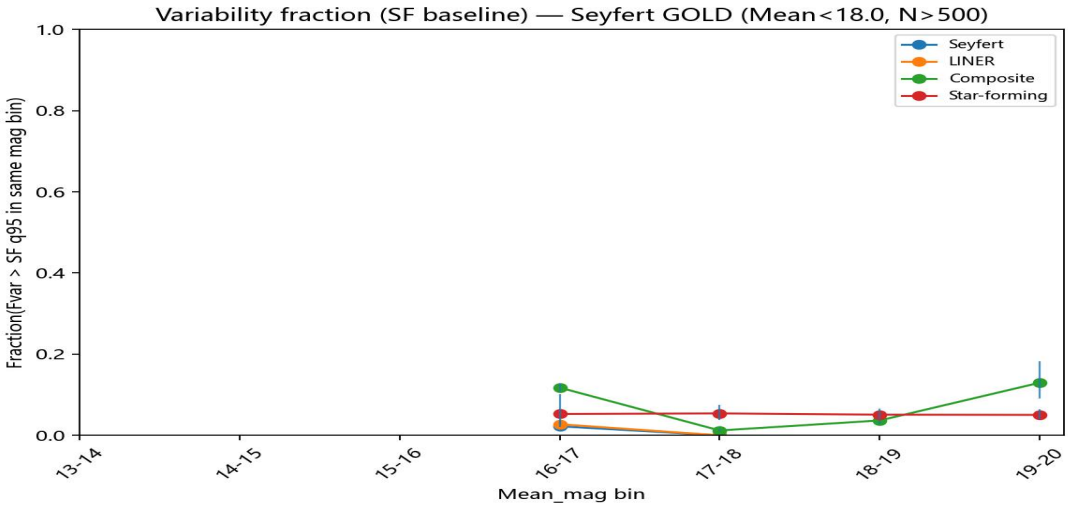

图 3 基于星等分箱的可变率：Seyfert GOLD（Mean<18，N>500）相对 Star-forming 噪声基线的对比

与图 2 相同，图 3 在每个平均星等分箱内使用 star-forming 样本的 Fvar 95%分位数作为噪声基线阈值，并计算各类别中满足 Fvar > SF_q95 的比例。不同之处在于：Seyfert 曲线仅使用 bright 的控制子样本（Mean_mag<18 且 N>500，Seyfert GOLD），以满足“先从高信噪、采样充分的 Seyfert 开始”的分阶段验证要求。该图用于评估在更受控的 Seyfert 样本中，是否存在相对于 SF 噪声基线的“excess variability”；误差棒表示分箱内比例的不确定度。

3.4 chi2red（含系统噪声地板 sys000–sys050）的显著性诊断

采用 chi2red_sysXXX > 1.5，并比较 sys000…sys050。结果显示显著性对系统噪声地板非常敏感，sys050 更保守、更接近合理范围。

交付物：

① Chi2redFraction_magbin_gt_1.5_sys000.png -- sys050.png

② coverage_by_sys_and_class.csv

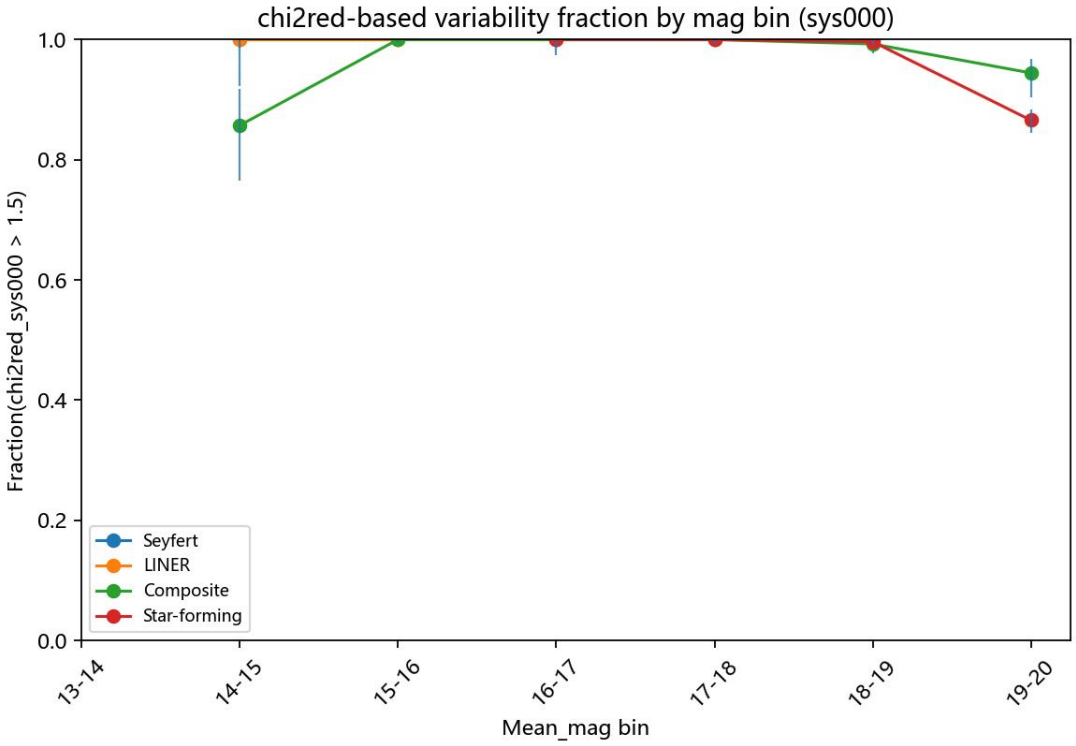

图 4 基于 χ²_red 的可变率（chi2red_sys000 > 1.5）：按星等分箱统计

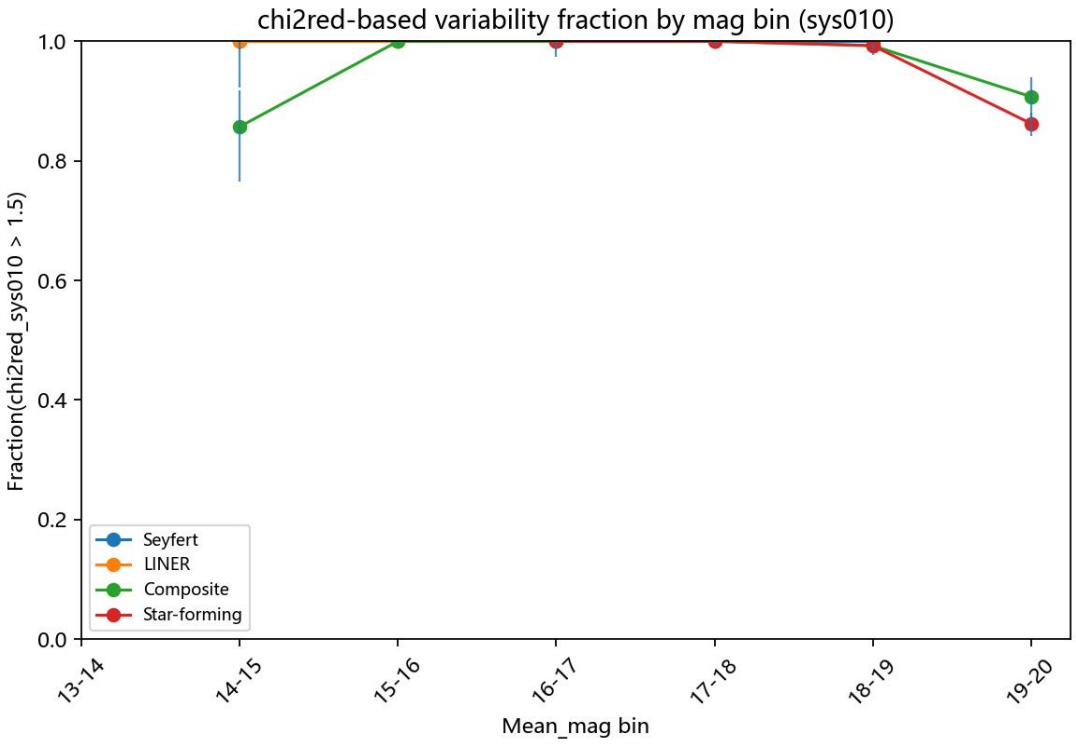

图 5 基于 χ²_red 的可变率（chi2red_sys010 > 1.5）：按星等分箱统计

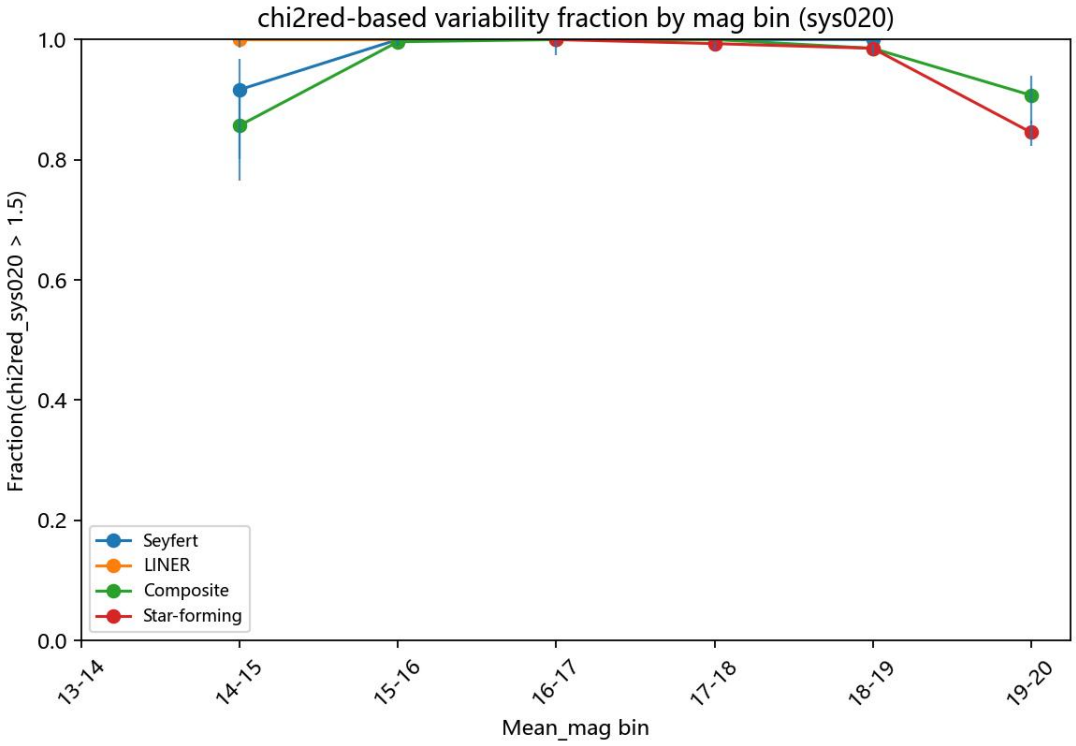

图 6 基于 χ²_red 的可变率（chi2red_sys020 > 1.5）：按星等分箱统计

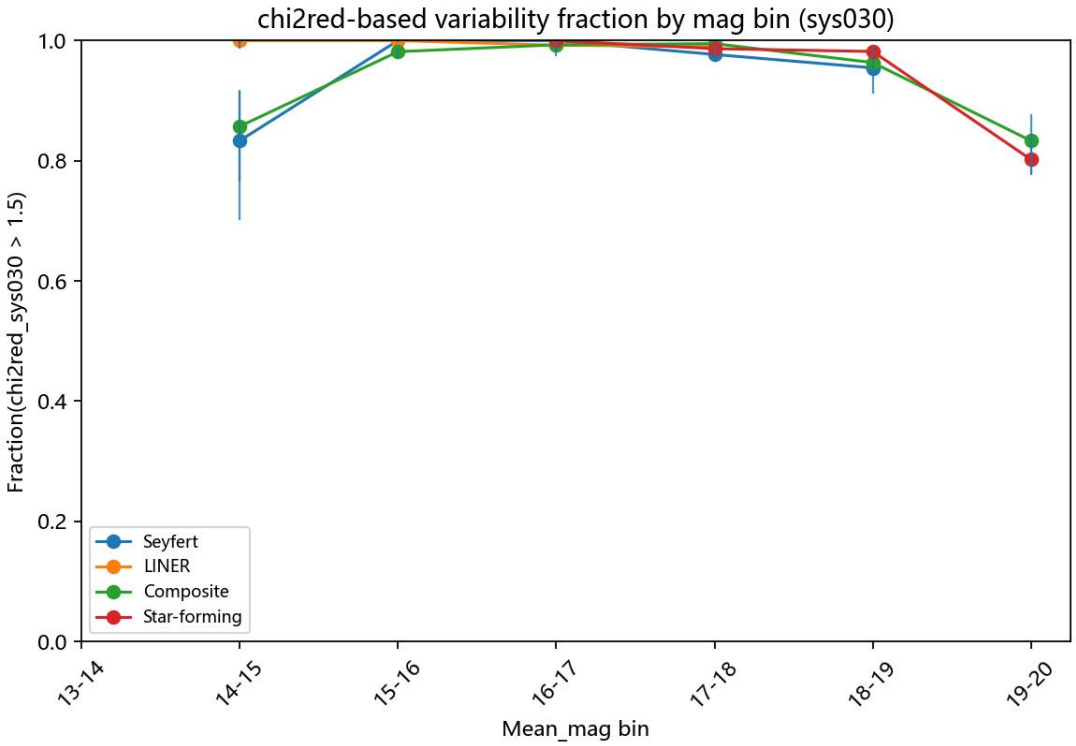

图 7 基于 χ²_red 的可变率（chi2red_sys030 > 1.5）：按星等分箱统计

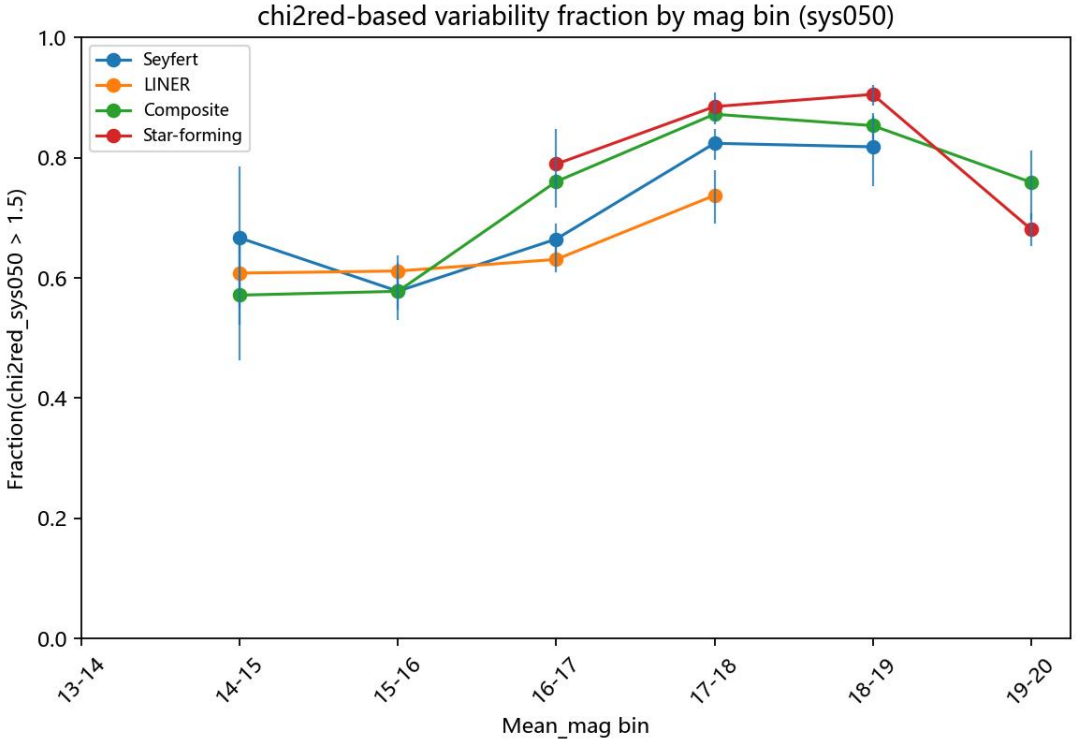

图 8 基于 χ²_red 的可变率（chi2red_sys050 > 1.5）：按星等分箱统计

为评估可变性显著性判据对测光系统误差假设的依赖程度，我们在 v7 统计表中采用一组系统误差地板（systematic floor）进行敏感性测试：sys000、sys010、sys020、sys030 与 sys050。对每个 sys 档位，我们计算对应的 reduced chi-square 指标$\chi^2_{red,sys}$，并在平均 r-band 星等分箱内统计各 BPT 类别（Seyfert、LINER、Composite、Star-forming）满足 $\chi^2_{red,sys} > 1.5$ 的比例。结果显示，当系统误差地板设得较小（sys000 – sys030）时，“显著可变比例”会在多数星等段接近 1，甚至 star-forming 样本也出现饱和，说明在偏乐观的误差模型下该显著性判据过于宽松；而在更保守的系统误差地板（sys050）下，比例不再饱和，呈现更为合理的范围与趋势，因而可作为更稳健的噪声、系统效应基准。

4. SF、Composite 污染复查

从 SF 与 Composite 各选 20 个 Fvar>10%对象，生成 QC 图册与汇总表。多数高 Fvar 对象表现出 near-limit 或较大 magerr 特征，提示噪声、系统误差可造成“假高变”；同时保留少数较干净的候选以供后续深入验证。

交付物：

① Starforming_highFvar_QC_booklet.pdf

② Composite_highFvar_QC_booklet.pdf

③ highFvar_qc_summary.csv

5. Outlier 处理策略

目前不进行 MAD 自动裁剪作为主流程；仅对 outlier 做标记并在 QC 图中展示，后续将在明确 outlier 来源后再讨论是否需要裁剪，避免误删真实 flare。

6.当前阶段核心结论

（1）MINFIX 下 bright Seyfert（Mean<18、N>500）QC 图册显示：r-band 单波段、oid_n=1、catflags=0，曲线整体物理可信，极端点仅标记不删。

（2）全样本中 Fvar 随 Mean_mag 变暗显著增大，说明暗端噪声、系统误差主导，因此固定阈值不再合适。

（3）采用 SF 在各 mag-bin 的 Fvar 95%分位作为“噪声基线阈值”，可将阈值随星等自适应调整，SF 自身可变率接近 5%。

（4）在该基线下，Seyfert GOLD 的可变率对比提供了更受控的“excess variability”评估框架。

（5）chi2red 显著性对系统噪声地板 sys000–sys050 敏感，sys050 更保守，提示误差、系统项建模仍是决定性因素。

（6）对 SF、Composite 的高 Fvar 对象逐个 QC 表明，多数高 Fvar 与 near-limit、大误差相关，支持其为噪声驱动的可能；少数相对干净候选将进入下一步深入验证。

（7）outlier 暂不自动裁剪，避免误删真实天体事件。

7.下一步计划

（1）在 Seyfert 控制样本上收敛“最终可变判据”（例如 SF-baseline + 保守 sys050 chi2red 的组合）。

（2）优先对“干净的高变候选”（n_near_limit 低、max_magerr 低、无明显批次台阶）做更深入验证。

（3）方法确认后再扩展到更弱 Seyfert，再到 LINER、Composite、SF 全样本。

The objective of this task is to first ensure that the photometric results are physically reliable, capable of distinguishing between noise, systematic effects, and genuine variations. The process will be validated on a small, controllable Seyfert sample before being gradually extended to fainter AGNs and other categories.

1. Data reading and sanity check

1.1 Uniqueness of the Core Source

The MINFIX r-band photometric curves used in this study all display oid_n=1 in the QC plot, and exclusively employ the r-band to eliminate the risk of 'potentially conflating multiple sources into a single curve' as previously observed.

1.2 Quality Control

The QC chart shows that all usage data meet the requirements:

(1)catflags=[0]

(2) Outliers are only marked, not automatically deleted

(3) The area near limitmag is explicitly marked to identify low SNR noise-dominant regions

deliverable：

Seyfert_bright_QC_booklet.pdf

2. Seyfert control sample (bright and well-sampled) QC

2.1 Sample Definition

(1) Mean_mag < 18

(2) N > 500

2.2 Representative Light-Duty Curve Atlas

Generate 30 high-quality (QC) images of bright Seyfert galaxies, with each page containing:

(1) Light curve (including error bars)

(2) Outliers are marked with red circles, and near-limits with orange circles.

(3) Metadata (Mean, Fvar, N, ΔT)

(4) Integrity indicators (oid_n, catflags, n_outlier, n_near_limit)

deliverable ：

① Seyfert_bright_QC_booklet.pdf

② qc_index.csv

3. Distinguishing true variation from noise

3.1 Evidence: The fixed threshold Fvar>5% is unreasonable

The Fvar of the whole sample increases significantly with the darkening of Mean_mag, and the noise and systematic error dominate the dark end.

Delivery image:

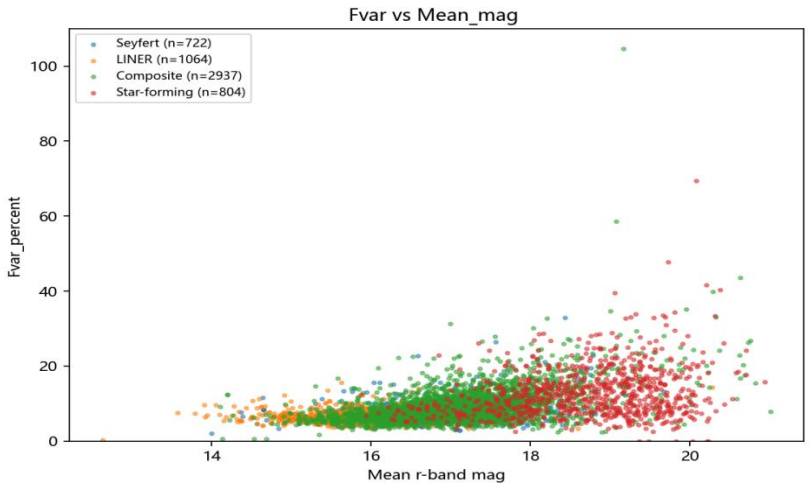

Figure 1 Fvar–mean magnitude relation of galaxies in different BPT categories (ZTF r-band)

Figure 1 illustrates how the fractional variation amplitude (Fvar Percent) of Seyfert, LINER, Composite, and star-forming samples in the ZTF r-band fluctuates with Mean r-band mag. The data reveals that the dispersion of Fvar increases significantly in dimmer sources, indicating that photometric errors and systematic effects dominate at the lower magnitude end. Consequently, using a fixed threshold (e.g., Fvar> 5%) may result in a higher proportion of "false variable" signals in the dimmer region. Future analyses should incorporate magnitude-dependent noise baselines or more robust significance metrics.

3.2 Establishing a "magnitude-dependent" noise baseline threshold using SF

Within each Mean_mag bin, the 95th percentile of Fvar (for star-forming) is used as the threshold instead of a fixed cut.

Key table:

SF_thresholds_by_magbin_Fvar.csv

3.3 Comparison of Variable Rates Based on SF Threshold

Export two images:

① Mean Magnitude bin of variable rate vs Mean Magnitude for all four categories

② Seyfert GOLD (Mean<18, N>500) version

Delivery image:

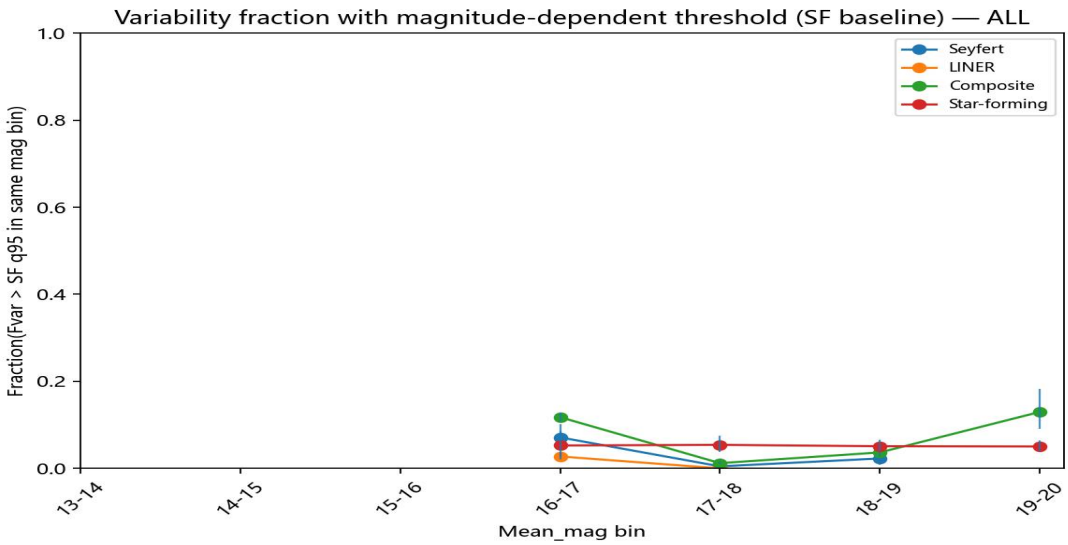

Figure 2 Variable rate based on magnitude binning: Star-forming noise baseline for star correlation thresholds (full sample)

Figure 2 illustrates the proportion of objects meeting Fvar> SF_q95 across different average magnitude bins for various categories (Seyfert, LINER, Composite, Star-forming). Here, SF_q95 denotes the 95th percentile of Fvar for star-forming samples within a given magnitude bin, serving as a baseline threshold for noise and systematic effects. Since the threshold is defined by SF, the proportion of SF itself in each bin should approach 5%. These magnitude-related thresholds replace fixed thresholds to mitigate misjudgments caused by dark-side noise amplification. Error bars indicate the uncertainty of the bin-proportion.

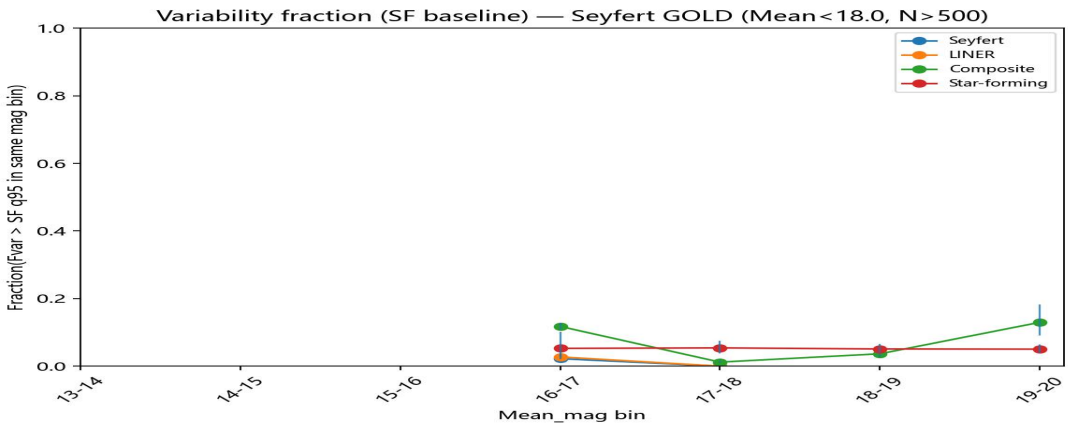

Figure 3 Comparison of variable rate based on magnitude binning: Seyfert GOLD (Mean<18, N>500) versus star-forming noise baseline

Similar to Figure 2, Figure 3 employs the 95th percentile of Fvar from star-forming samples as the noise baseline threshold within each mean magnitude bin, calculating the proportion of categories where Fvar exceeds SF_q95. The key distinction lies in Seyfert curves, which exclusively use the bright control subset (Mean_mag<18 and N>500, Seyfert GOLD) to meet the staged validation requirement of "starting with high SNR, well-sampled Seyfert objects." This figure evaluates the presence of "excess variability" relative to the SF noise baseline in more controlled Seyfert samples, with error bars indicating the uncertainty of the bin-proportion estimates.

3.4 Chi-square test for red noise (including system noise floor sys000 –sys050)

Using chi2red_sysXXX> 1.5 and comparing sys000...sys050, the results show that significance is highly sensitive to the system noise floor, with sys050 being more conservative and closer to the reasonable range.

deliverable ：

① Chi2redFraction_magbin_gt_1.5_sys000.png -- sys050.png

② coverage_by_sys_and_class.csv

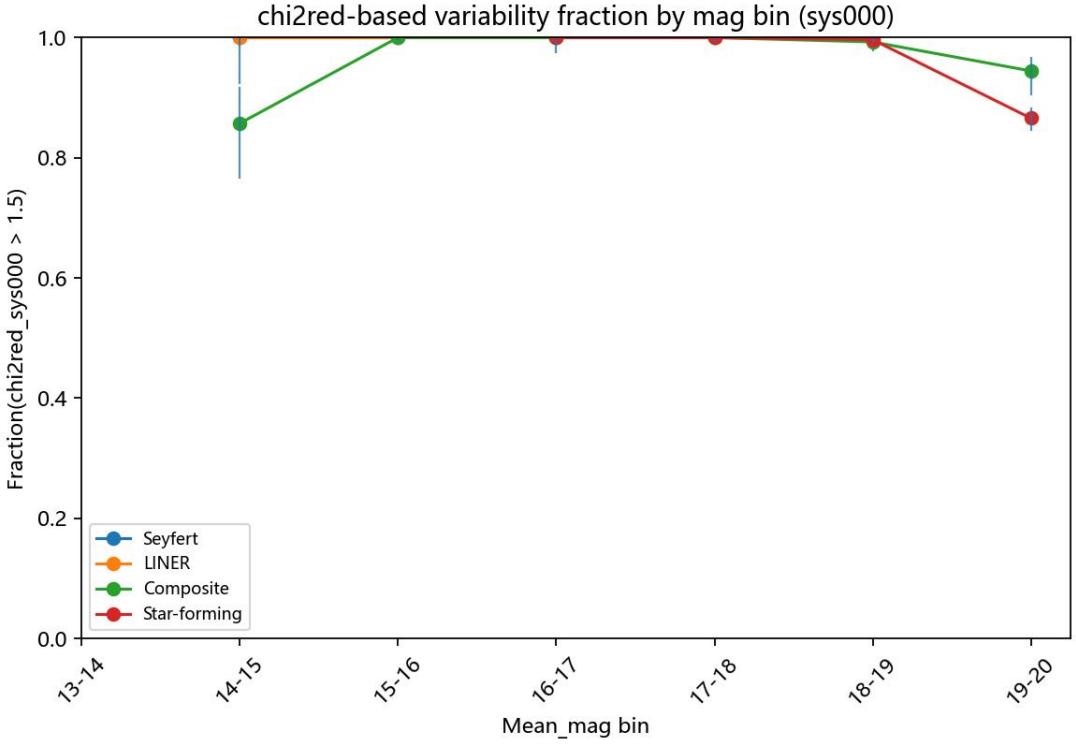

Figure 4: Variable rate based on χ^2_red (chi2red_sys000> 1.5): binning by magnitude

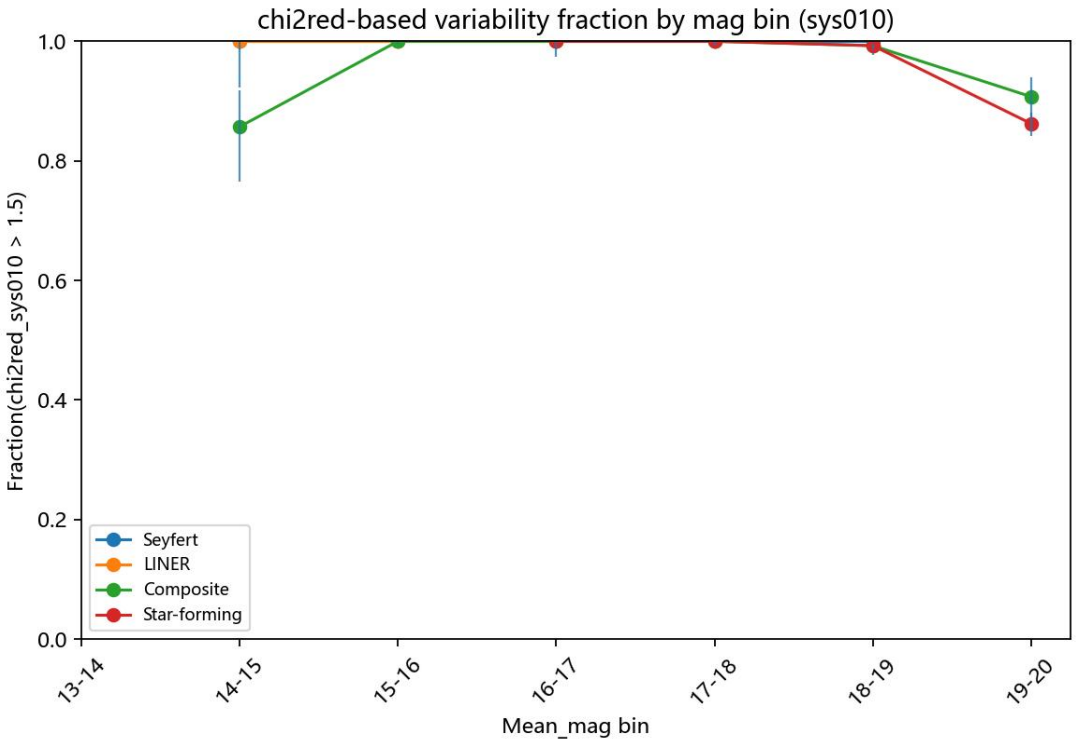

Figure 5: Variable rate based on χ^2_red (chi2red_sys010> 1.5): binning by magnitude

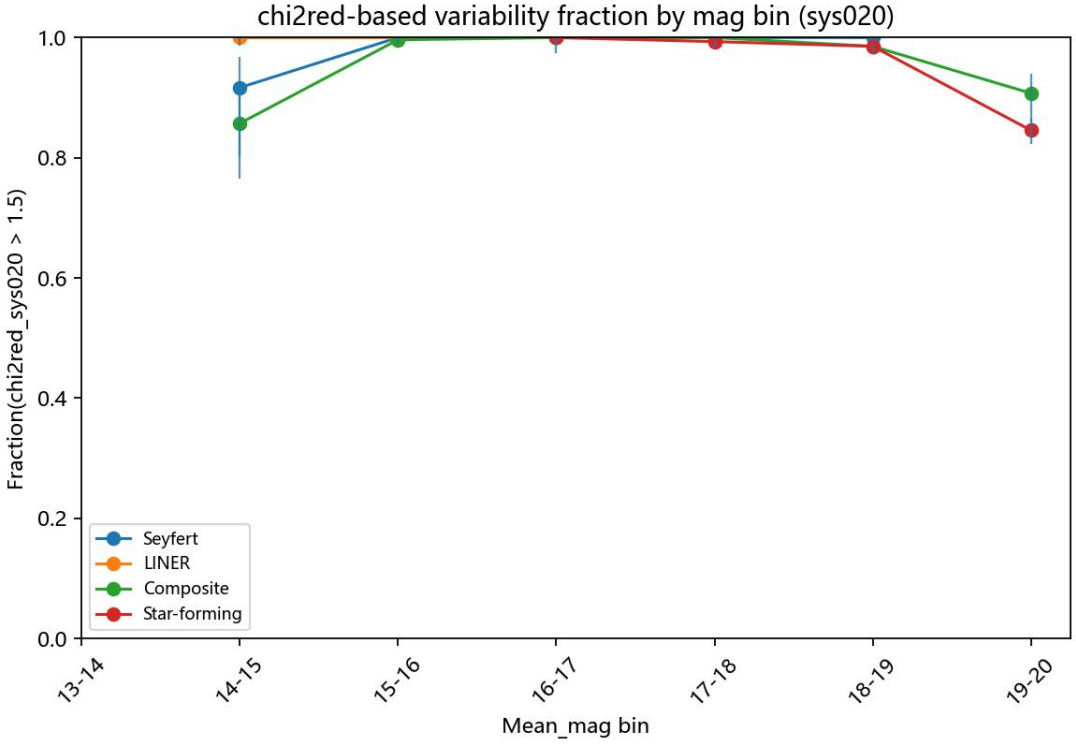

Figure 6: Variable rate based on χ^2_red (chi2red_sys020> 1.5): binning by magnitude

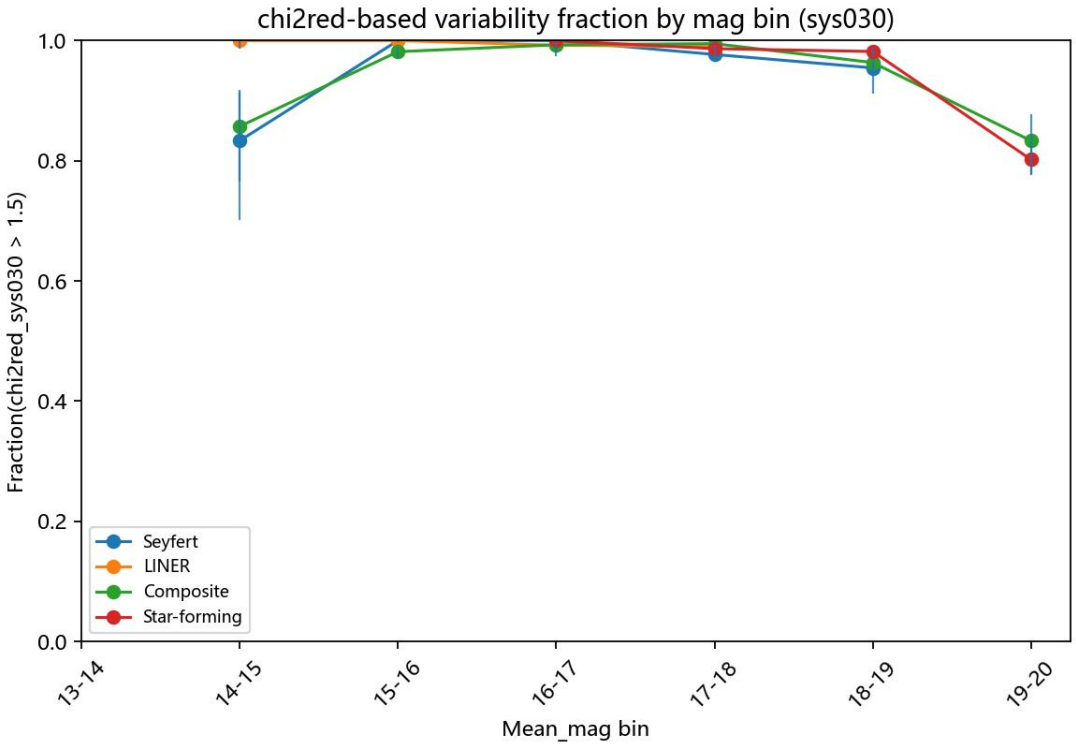

Figure 7: Variable rate based on χ^2_red (chi2red_sys030> 1.5): binning by magnitude

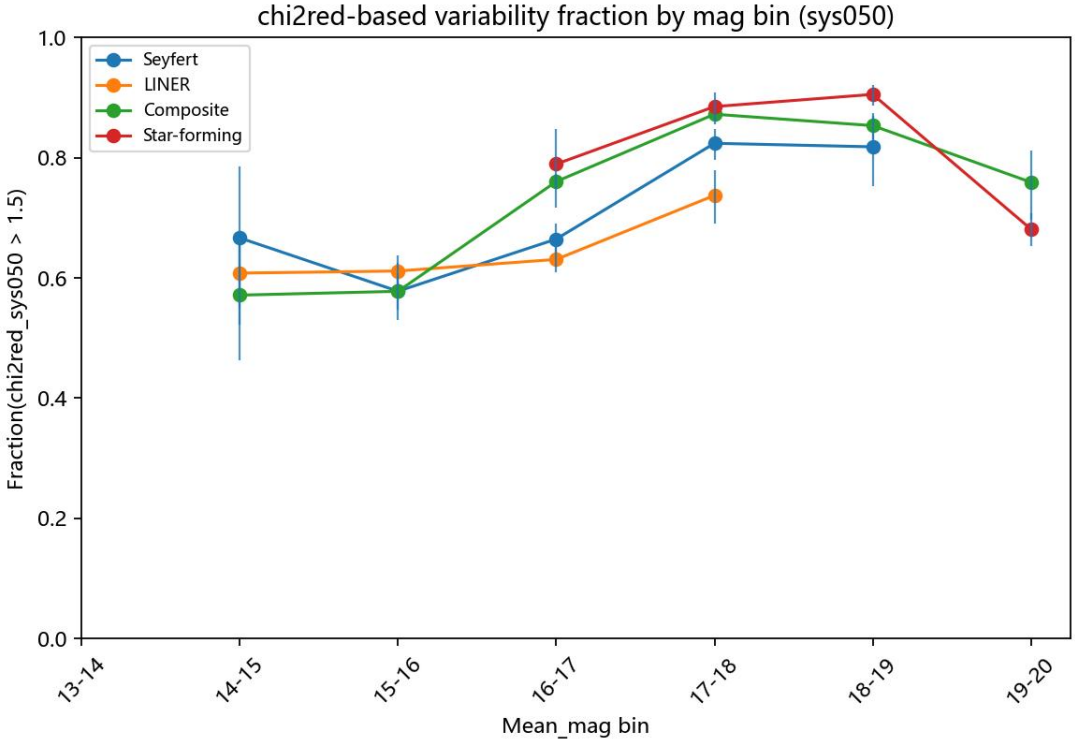

Figure 8: Variable rate based on χ^2_red (chi2red_sys050> 1.5): binning by magnitude

To evaluate the dependence of variability significance criteria on the assumption of photometric system errors, we conducted sensitivity tests using a set of systematic floor values (sys000, sys010, sys020, sys030, and sys050) in the v7 statistical table. For each sys level, we calculated the corresponding reduced chi-square index $\chi^2_{red,sys}$ and statistically analyzed the proportion of BPT categories (Seyfert, LINER, Composite, Star-forming) that satisfy $\chi^2_{red,sys} > 1.5$ within the average r-band magnitude bins. The results showed that when the systematic floor was set to lower values (sys000−sys030), the "significant variability proportion" approached 1 across most magnitude bands, even saturating in star-forming samples, indicating that the significance criterion was overly lenient under a biasedly optimistic error model. In contrast, with the more conservative systematic floor (sys050), the proportion no longer saturated, demonstrating a more reasonable range and trend, thus serving as a robust benchmark for noise and systematic effects.

4. SF, Composite pollution re-examination

Select 20 objects with Fvar> 10% from both SF and Composite datasets to generate QC catalogs and summary tables. Most high-Fvar objects exhibit near-limit or large-magerr characteristics, indicating that noise or systematic errors may cause "false high variations"; meanwhile, a small number of cleaner candidates are retained for subsequent in-depth validation.

deliverable ：

① Starforming_highFvar_QC_booklet.pdf

② Composite_highFvar_QC_booklet.pdf

③ highFvar_qc_summary.csv

5. Outlier Handling Strategy

MAD automatic clipping is not currently the primary workflow. Outliers are only flagged and displayed in the QC chart. We will evaluate clipping necessity after identifying the source of outliers to prevent accidental removal of genuine flare data.

6. Core Conclusions of the Current Phase

(1) The MINFIX bright Seyfert (Mean<18, N>500) QC catalog shows: for r-band single-band with oid_n=1 and catflags=0, the curve is physically credible overall, with only extreme points marked but not deleted.

(2) The Fvar in the whole sample increases significantly with Mean_mag darkening, indicating that the dark end noise and systematic error dominate, so the fixed threshold is no longer appropriate.

(3) The 95% Fvar percentile of SF in each mag-bin is used as the "noise baseline threshold", which can be adjusted adaptively with magnitude, and the variable rate of SF itself is close to 5%.

(4) Under this baseline, the variable rate comparison in Seyfert GOLD provides a more controlled framework for assessing excess variability.

(5) The chi2red significance test indicates that system noise floor sys000–sys050 is sensitive, with sys050 being more conservative, suggesting that error and system term modeling remain decisive factors.

(6) QC of high Fvar objects in SF and Composite shows that most high Fvar values are associated with near-limit and large errors, supporting the possibility of noise-driven effects; a few relatively clean candidates will proceed to further validation.

(7) Outliers are not automatically trimmed to prevent accidental deletion of actual celestial events.

7. Next steps

(1) Convergence of the "final variable criterion" (e.g., the combination of SF-baseline and conservative sys050 chi2red) on the Seyfert control sample.

(2) Prioritize in-depth validation of "clean high-energy candidates" (characterized by low n_near_limit, low max_magerr, and absence of distinct batch steps).

(3) After method validation, the approach was extended to weaker Seyfert systems, then to LINER, Composite, and SF full-samples.